Assessing Risk: Competition & Data

Module 4

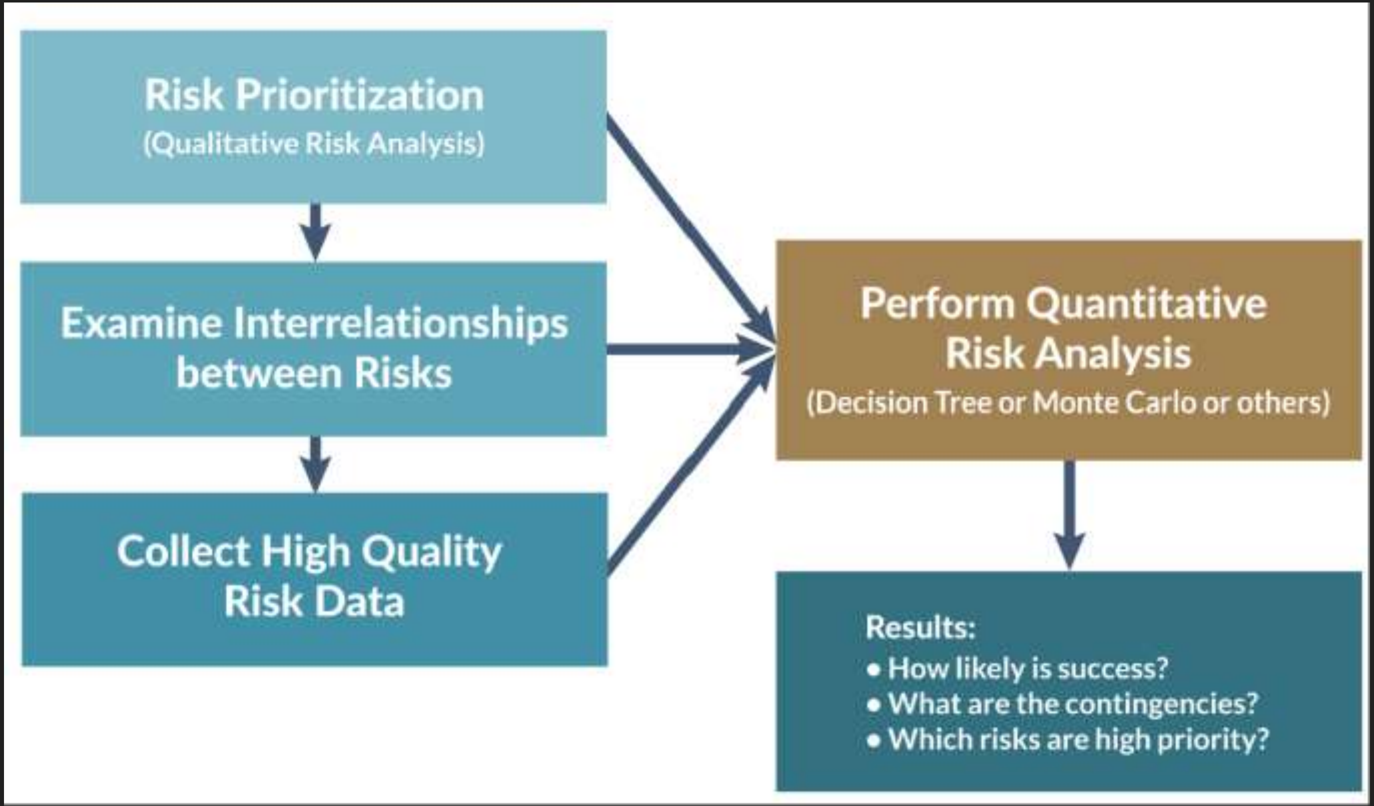
Objectives

- Quantitative risk analytics methods and techniques in risk assessment
- Develop risk mitigation measures

Quantitative analysis answers:

- What is the likelihood or probability of meeting the organization's objectives?
- How much contingency planning or mitigation actions are needed?
- What individual risks contribute to overarching risk to the organization?

Process of Quantitative Risk Analysis



Quantitative Risk Assessment Techniques

Uses statistical models, simulation, and ML to estimate:

- Probability of risk events occurring
- Magnitude of potential impact
- Optimal mitigation strategies

Quantitative Risk Assessment Techniques

Data Sources for Market Entry Analysis

Structured Data

- Competitor entry history (market, region, entry year, outcomes)
- Macroeconomic indicators (GDP, interest rates, inflation)
- Market metrics (market share, pricing, sales volume)
- Operational data (customer churn, cost structure, supply chain performance)

Quantitative Risk Assessment Techniques

Data Sources for Market Entry Analysis

Unstructured Data (Qualitative)

- News articles
- Press releases
- Earnings call transcripts
- Social media sentiment
- Expert reports and industry whitepapers

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Quantitative Modeling Techniques

Statistical Risk Models: relationships between risk drivers and outcomes.

Examples:

- Logistic regression for probability of competitor success
- Linear regression for impact on revenue
- Time-series models for market trend forecasting

Libraries:

- scikit-learn
- statsmodels
- Prophet (Meta)

Quantitative Modeling Techniques

Monte Carlo Simulation: Simulates thousands of scenarios to model uncertainty.
- Revenue impact of new competitors
- Pricing strategy risks
- Supply chain disruptions

Libraries:
- NumPy
- SciPy
- Chaospy
- SimPy

Quantitative Modeling Techniques

ML Risk Prediction: to predict future risk events.

- Models: Random Forest, Gradient Boosting, Neural Networks
- Predicting market entry success
- Estimating market share loss
- Detecting early warning signals

Libraries: scikit-learn, XGBoost, LightGBM, CatBoost

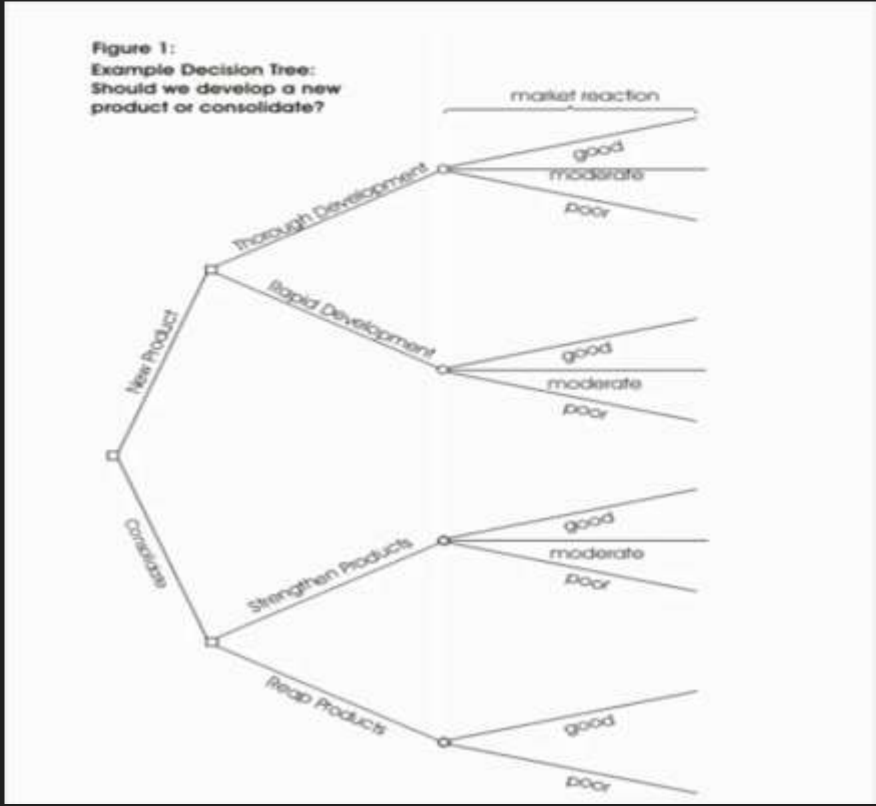
Rirk Visualization and Decision Support

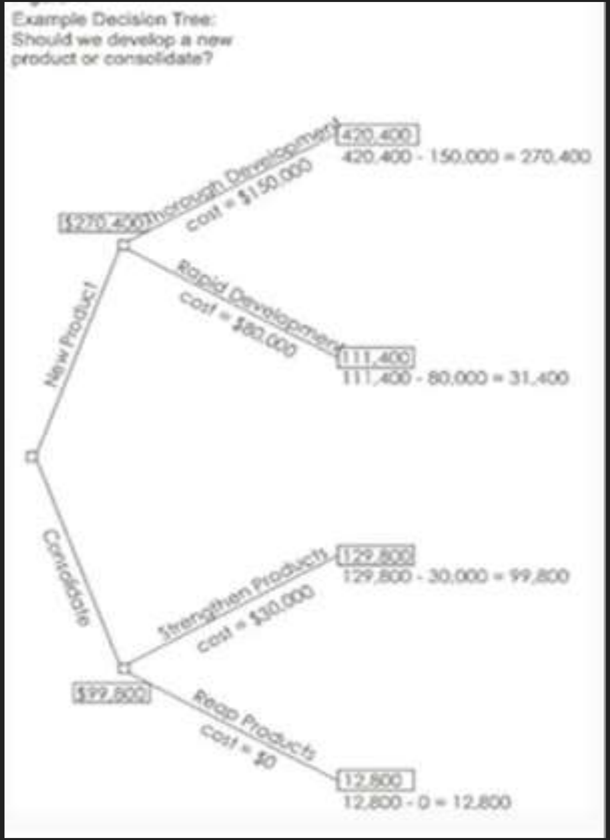
- Risk heat maps
- Scenario comparison charts
- Probability distributions

Quantitative Risk Assessment Techniques

- 3-Point Estimate- optimistic, most likely and pessimistic estimates of risk cost.
- Decision Tree Analysis- presents costs associated with different choices with respect to risk.
- Expected Monetary Value (EMV)- calculate a contingency reserve for a project budget based on expected cost associated with opportunities and threats of the project .
- Monte Carlo Simulation
- Sensitivity Analysis
- Fault Tree Analysis

Decision Tree Analysis





Expected Monetary Value (EMV)

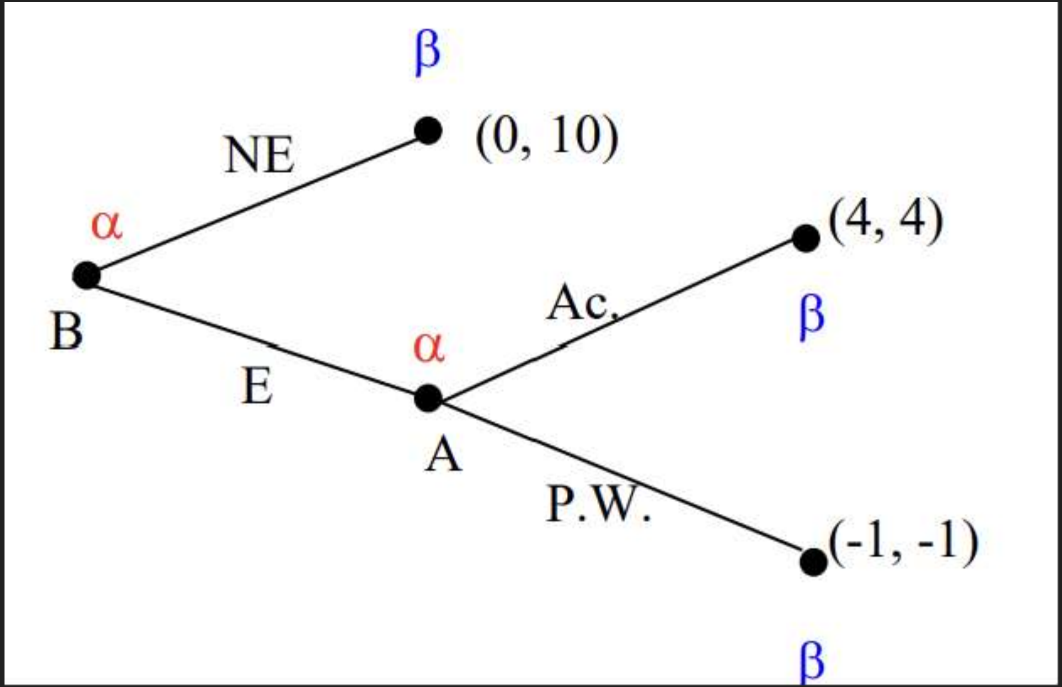
- Determine a contingency reserve for a project.

  EMV = Probability X Cost Impact

Risk	Probability	Cost Impact	EMV
A (Threat)	30%	$70,000	$21,000
B (Threat)	10%	$40,000	$4,000
C (Opportunity)	50%	($20,000)	($10,000)
		Total EMV	$15,000

Decision Tree Analysis

Market entry game with imperfect information



Risk management techniques

- Acceptance
- Mitigation
- Avoidance
- Transfer

Output of Quantitative Risk Analysis

- Probability distributions of outcomes
- Scenario simulations
- Expected loss estimates
- Risk-adjusted decision recommendations

Note

- Assess risks based on the controls
- Risk mitigating is evolutionary